Let A be a square matrix of size $n \times n$.

$\boxed{\text{Case : } n = 2}$

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

The determinant of A is a real number given by

$$\det(A) = |A| = a_{11}a_{22} - a_{21}a_{12}$$

Examples

1) $A = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$

$$\det(A) = \begin{vmatrix} 1 & 2 \\ 3 & 5 \end{vmatrix} = 1(5) - 3(2) = -1$$

2) $B = \begin{bmatrix} 2a & -1 \\ -1 & 3a \end{bmatrix}$

$$\det(B) = \begin{vmatrix} 2a & -1 \\ -1 & 3a \end{vmatrix} = 2a(3a) - (-1)(-1)$$

$$= 6a^2 - 1$$

3) For which values of a, we have $\det(A) = 0$ where $A = \begin{bmatrix} a-2 & 1 \\ -5 & a+4 \end{bmatrix}$.

$$\begin{vmatrix} a-2 & 1 \\ -5 & a+4 \end{vmatrix} = (a-2)(a+4) - (-5)(1)$$

$$= a^2 + 2a - 8 + 5$$

$$= a^2 + 2a - 3$$

$$\det(A) = 0 \iff a^2 + 2a - 3 = 0$$

$$\iff (a-1)(a+3) = 0$$

$$\iff a = 1 \text{ or } a = -3$$

$\boxed{\text{Case : } n = 3}$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$\begin{array}{ccc} + & - & + \end{array}$$

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11}\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12}\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13}\begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

It is better to expand the determinant from the row or column that has a lots of zeros.

Example

$$\begin{vmatrix} 1 & 2 & 3 \\ -1 & 0 & 1 \\ 3 & 5 & 4 \end{vmatrix} = -(-1)\begin{vmatrix} 2 & 3 \\ 5 & 4 \end{vmatrix} + 0 - 1\begin{vmatrix} 1 & 2 \\ 3 & 5 \end{vmatrix}$$

$$= 1(8 - 15) - (5 - 6)$$

$$= -7 + 1 = -6$$

Sarus method : can be applied only for matrices of size 3x3.

$$\left|\begin{array}{ccc|cc} a_{11} & a_{12} & a_{13} & a_{11} & a_{12} \\ a_{21} & a_{22} & a_{23} & a_{21} & a_{22} \\ a_{31} & a_{32} & a_{33} & a_{31} & a_{32} \end{array}\right.$$

$$= a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32}$$

$$- a_{13}a_{22}a_{31} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33}$$

Example

$$\begin{array}{c} + \\ - \\ + \end{array}\begin{vmatrix} a & b & 0 \\ 0 & a & b \\ a & 0 & b \end{vmatrix} = a\begin{vmatrix} a & b \\ 0 & b \end{vmatrix} - 0 + a\begin{vmatrix} b & 0 \\ a & b \end{vmatrix}$$

$$= a(ab - 0) + a(b^2 - 0)$$

$$= a^2b + ab^2$$

$$\left|\begin{array}{ccc|cc} a & b & 0 & a & b \\ 0 & a & b & 0 & a \\ a & 0 & b & a & 0 \end{array}\right. = a^2b + b^2a + 0 - 0 - 0 - 0$$

$$= a^2b + b^2a$$

Example Find a so that $\det A = 0$ where

$$A = \begin{bmatrix} a-4 & 0 & 0 \\ 0 & a & 2 \\ 0 & 3 & a-1 \end{bmatrix}$$

$$\det A = \begin{array}{c} + \\ - \\ + \end{array}\begin{vmatrix} a-4 & 0 & 0 \\ 0 & a & 2 \\ 0 & 3 & a-1 \end{vmatrix}$$

$$= (a-4)\begin{vmatrix} a & 2 \\ 3 & a-1 \end{vmatrix} - 0 + 0$$

$$= (a-4)(a^2 - a - 6)$$

$$= (a-4)(a+2)(a-3)$$

$$\det A = 0 \iff a \in \{4, -2, 3\}$$

$\boxed{\text{Case } n > 3}$

general, a determinant Δ_n of a matrix A of size $n \times n$ can be evalued by the same method in terms of determinants of matrices of size $(n-1) \times (n-1)$, and respecting the alternating signs :

$$\begin{vmatrix} + & - & + & - & \cdots \\ - & + & - & + & \cdots \\ + & - & & & \\ \vdots & \vdots & & & \end{vmatrix}$$

Example

$$\begin{array}{c} + \\ - \\ + \\ - \end{array}\begin{vmatrix} 1 & 2 & -1 & 3 \\ 0 & 1 & 4 & 5 \\ 1 & -1 & 1 & -1 \\ 0 & 0 & 1 & 5 \end{vmatrix} = 1\begin{vmatrix} 1 & 4 & 5 \\ -1 & 1 & -1 \\ 0 & 1 & 5 \end{vmatrix} + 1\begin{vmatrix} 2 & -1 & 3 \\ 1 & 4 & 5 \\ 0 & 1 & 5 \end{vmatrix}$$

$$= 21 + 38 = 59$$

$$\left|\begin{array}{ccc|cc} 1 & 4 & 5 & 1 & 4 \\ -1 & 1 & -1 & -1 & 1 \\ 0 & 1 & 5 & 0 & 1 \end{array}\right. = 0 + 1 + 20 = 21$$

$$\left|\begin{array}{ccc|cc} 2 & -1 & 3 & 2 & -1 \\ 1 & 4 & 5 & 1 & 4 \\ 0 & 1 & 5 & 0 & 1 \end{array}\right. = 40 + 3 - 10 + 5 = 38$$